

CSC1042 Computer Network II

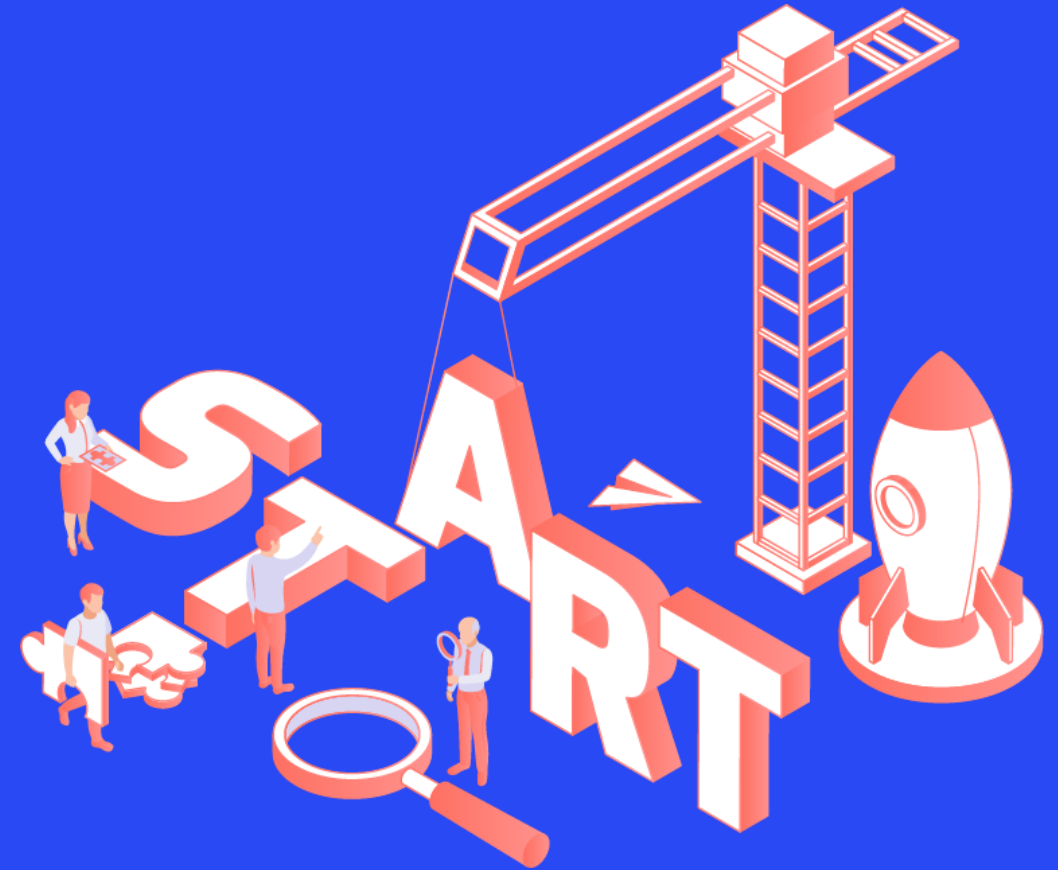
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**Numeral
systems**



learning Outcomes

- Understand different numbering systems
- Convert from decimal to binary and vice-versa
- Convert from decimal to hexadecimal and vice-versa



Decimal system

- Decimal system is base 10
 - 10 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- Using the decimal system, 2168 can be written as:

$$2168 = (2 \times 1000) + (1 \times 100) + (6 \times 10) + (8 \times 1)$$

$$= (2 \times 10^3) + (1 \times 10^2) + (6 \times 10^1) + (8 \times 10^0)$$

- This called positional notation
- Any decimal number can be written using this notation

	10^6	10^5	10^4	10^3	10^2	10^1	10^0
2168				2	1	6	8
67254			6	7	2	5	4

Binary system

- Binary system is base 2
 - Only 2 digits: 0 and 1
- Let's write the decimal number 168 as the sum of multiples of 2

$$168 = 128 + 32 + 8$$

$$= 1 \times 2^7 + 1 \times 2^5 + 1 \times 2^3$$

$$= 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$

- Therefore, 168 in binary is 10101000

	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
	128	64	32	16	8	4	2	1
168	1	0	1	0	1	0	0	0
224	1	1	1	0	0	0	0	0

Binary to Decimal

- Assume we have the binary number 01101101. What's this number in decimal?
- $01101101 = 0 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
0	1	1	0	1	1	0	1

$$= 0 + 64 + 32 + 0 + 8 + 4 + 0 + 1$$

$$= 109$$

Hexadecimal system

- Hexadecimal system is base 16
 - 16 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
 - Uses powers of 16

16^4	16^3	16^2	16^1	16^0
65536	4096	256	16	1

- Multiplying or dividing by 16 is not straightforward when doing conversion from hexadecimal to decimal and vice-versa

Hexadecimal, Decimal, and Binary equivalents

Hexadecimal
0
1
2
3
4
5
6
7
8
9
A
B
C
D
E
F

Decimal
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

Binary
0000
0001
0010
0011
0100
0101
0110
0111
1000
1001
1010
1011
1100
1101
1110
1111

- Each hex digit represents a 4-bit binary number
- We can express an 8-bit binary number using 2 hex digits

Hexadecimal to Decimal

- Assume we have the hexadecimal number 2A. What's this number in decimal?
- First convert 2A to binary
 - 2 in 4-bit representation: 0010
 - A (10 in decimal) in 4-bit representation: 1010
- Put the two representations side by side

2^3	2^2	2^1	2^0
8	4	2	1
0	0	1	0

2^3	2^2	2^1	2^0
8	4	2	1
1	0	1	0

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
0	0	1	0	1	0	1	0

$$32 + 8 + 2 = 42$$

Decimal to Hexadecimal

- Let's convert 197 in decimal to hexadecimal
- First convert 197 to binary
- Divide 11000101 into two 4-bits representations: 1100 and 0101
- Convert into decimal and then to its equivalent in hexadecimal

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
1	1	0	0	0	1	0	1

2^3	2^2	2^1	2^0
8	4	2	1
1	1	0	0

$8 + 4 = 12$
Hex C

2^3	2^2	2^1	2^0
8	4	2	1
0	1	0	1

$4 + 1 = 5$
Hex 5

- Therefore 197 in hexadecimal is C5 or 0xC5
 - 0x indicates that what follows is written in hexadecimal

THANK YOU